Daniel Fridljand

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Summary

Software consultant at TNG building production agentic AI systems. Sole AI engineer on a cinema-ticketing SaaS support-automation system, live in production since April 2026. Research background spans ETH Zürich (computational oncology), Stanford School of Medicine (first co-author in *Nature Medicine*, 2024), and EMBL (statistical genomics). M.Sc. Mathematics with full marks; Yale exchange scholar.

Software Development Experience

Software Consultant - Applied AI *Munich, Germany*
TNG Technology Consulting *Dec 2025 – present*

- **Email-to-Order Parsing (May 2026 – present)**: Designing a field-centric email-processing pipeline with FastAPI ingestion, Streamlit review dashboard, and attachment-aware multimodal LLM processing (text extraction, vision fallbacks for scanned PDFs) with parallel candidate resolution.
- **Customer Support Automation (Dec 2025 – Apr 2026, live in production)**: Sole AI engineer end-to-end on a cinema-ticketing SaaS — 175 live tickets processed at 72.9% strict / 81.3% content-supported approval over a two-week observation window; daily volume scaled 5x after CEO showcase.
- Architected a hybrid deterministic + agentic resolution workflow over 19 customer-intent categories using **Temporal** for durable orchestration with human-in-the-loop approval via Signals, and **DSPy/GEPA** for prompt optimization.
- Built a production evaluation suite in **Langfuse** (6 evaluation types, 8 score metrics, 14-label outcome taxonomy, LLM-as-judge) driving 13+ feature improvements directly from live reviewer feedback.
- Designed a reproducible 22-stage **Snakemake** pipeline (Microsoft Presidio PII detection, balanced sampling, automated LLM labeling) with GDPR-compliant censored/uncensored data paths.
- Tech: Python, FastAPI, Streamlit, PydanticAI, Temporal, Langfuse, Snakemake, Docker, DSPy, MCP.

Software Consultant - Enterprise Modernization *Munich, Germany*
TNG Technology Consulting *Dec 2024 – Dec 2025*

- Modernized a mission-critical global supply-chain application (Java 8 → 17, JBoss → WildFly) as part of the platform team in a multi-year transformation program.
- Shipped a JFrog Artifactory proxy in 3 days, reducing a recurring CI pipeline runtime from 8 hours to 30 seconds — saving developers ~1–2 hours per week each.
- Established DevSecOps governance: OWASP scans in CI, Grafana CVE dashboards, and SOAP→REST migration with Keycloak; architected daemonless Docker/Podman build environments.

Data Science Experience

Research Data Analyst - Computational Oncology *Basel, Switzerland*
ETH Zürich - Department of Biosystems Science and Engineering *Feb 2024 – Sept 2024*

- Developed novel Bayesian non-parametric methods (Hierarchical Dirichlet Process) for mutational signature estimation, extended to incorporate phylogenetic tree structures.
- Analyzed single-cell whole-exome sequencing data from the Tumor Profiler Study (187 cells, 10 melanoma tumors), identifying eight latent mutational signatures.
- Implemented hierarchical dependency structures in R where signature distributions for child phylogenetic nodes are drawn from parent distributions, enforcing biological inheritance.

Research Data Analyst - Environmental Epidemiology *Palo Alto, USA*
Stanford University School of Medicine *July 2023 – Dec 2023*

- **Nature Medicine (2024)**: Led the statistical analysis as first co-author, quantifying air pollution's contribution to racial and socioeconomic mortality disparities in the US.
- Engineered a big-data pipeline harmonizing 63+ million death records, satellite pollution estimates, and census demographics across 3,000+ US counties (1990–2016).

- Implemented confounder-adjusted causal inference (DAGs, propensity scoring), revealing >50% of the Black–White all-cause mortality difference is attributable to environmental factors.
- Built an R Shiny analytical application used directly by epidemiologists and policy researchers to explore 17-dimensional data.

Research Data Analyst - Statistical Genomics (Master's Thesis) — *Heidelberg, Germany*
European Molecular Biology Laboratory (EMBL) — *Oct 2021 – May 2022*

- Developed **IHW-Forest**, a novel multiple-testing method using Random Forests for hypothesis weighting, increasing discovery power by >30% on 16 billion genetic association tests.
- Optimized core splitting and weighting logic in C++ via Rcpp for high-performance processing of large-scale genomic data.
- Presented at seven scientific events including Yale University and a competitively selected oral contribution at DAGStat 2022.

Education

University of Heidelberg — *Heidelberg, Germany*
M.Sc. in M.Sc. Mathematics (Top of Class) — *Oct 2020 – May 2023*

- Grade: 1.0 (summa cum laude). Focus: ML, High-Dimensional Statistics, Probability Theory.
- Yale Exchange Scholar (2022–23, Honors): Deep Learning; Geometric & Topological ML (Prof. Krishnaswamy); Differentiable Manifolds.
- Master's Thesis (Grade 1.0) at EMBL: 'Better multiple Testing: Using multivariate co-data for hypothesis weighting'.

Publications

Disparities in air pollution attributable mortality in the US population by race, ethnicity and sociodemographic factors — July 2024
Pascal Geldsetzer, ***Daniel Fridljand***, Mathew V. Kiang, Eran Bendavid, Sam Heft-Neal, Marshall Burke, et al.
www.nature.com/articles/s41591-024-03117-0 (Nature Medicine)

Projects

Data Mining Hackathon - Core Demand Prediction (TUM.ai x Unite) — *Online*, *Mar 2026*

- **3rd in Level 2** (net €423K) and **5th in Level 1** (net €1.33M) with a two-stage LightGBM model (recurrence classifier + spend regressor) combined via an expected-utility objective.
- Engineered 200+ features and a reproducible Snakemake pipeline with automated portal submission via Playwright.

Agent Olympics Hackathon Munich 2026 — *Munich*, *Jan 2026*

- Engineered a white-hat voice agent executing automated prompt-injection attacks across email, SMS, and WhatsApp, with automated OSINT-driven persona profiling.

Skills

Programming: Python (pandas, FastAPI, PyTorch, PydanticAI, DSPy), R (tidyverse, Shiny, Rcpp, Bioconductor), Java 17 (Spring), TypeScript/React, C++, SQL.

Applied AI: Agentic systems, LLM orchestration (PydanticAI, DSPy, MCP), LLM evaluation (Langfuse), prompt engineering, Causal Inference, Bayesian Non-parametrics.

Infrastructure: Docker, Podman, Kubernetes, CI/CD (Jenkins, GitHub Actions), AWS, Temporal, Snakemake, Prometheus, Grafana.

Languages: English (C2), German (Native), Russian (Native).